6. Close the loop from the digital environment back into the physical system via some form of action.

Over the course of this book, we will explore each of these components in greater detail. We will work through an example of how you can build a functioning digital twin of perhaps your most important asset—your home—and see how building a digital representation can provide value to you in your everyday life. While a home digital twin might seem modest compared to industrial applications, the same architectural patterns, software components, and data flows apply directly to factories, power plants, and smart cities.

1.5.1 A clear definition of success

Before writing a single line of code or creating any rich interactive visualizations, you should be able to clearly articulate what you hope to achieve by building a digital twin. In early stages, this might mean establishing a systematic approach to data collection, identifying relationships between variables, and gaining a clearer understanding of system behavior. These exploratory outcomes provide the foundation for defining more specific targets.

As your digital twin matures, success should be defined through measurable outcomes that drive behavioral change, not just numerical outputs. Rather than vague aspirational priorities like "improving operations" or "reducing cost", articulate how the digital twin will change decisions and actions. For example, "reduce the London hotel’s electricity consumption by 10% within 6 months by enabling facility managers to optimize HVAC control based on real-time occupancy data" is a goal that combines quantifiable results with the behavioral shift that achieves them.